

**I LEFT SSH OPEN FOR AN
HOUR.**

**Then I analyzed the packets that
were thrown at me.**

Swipe to see how →

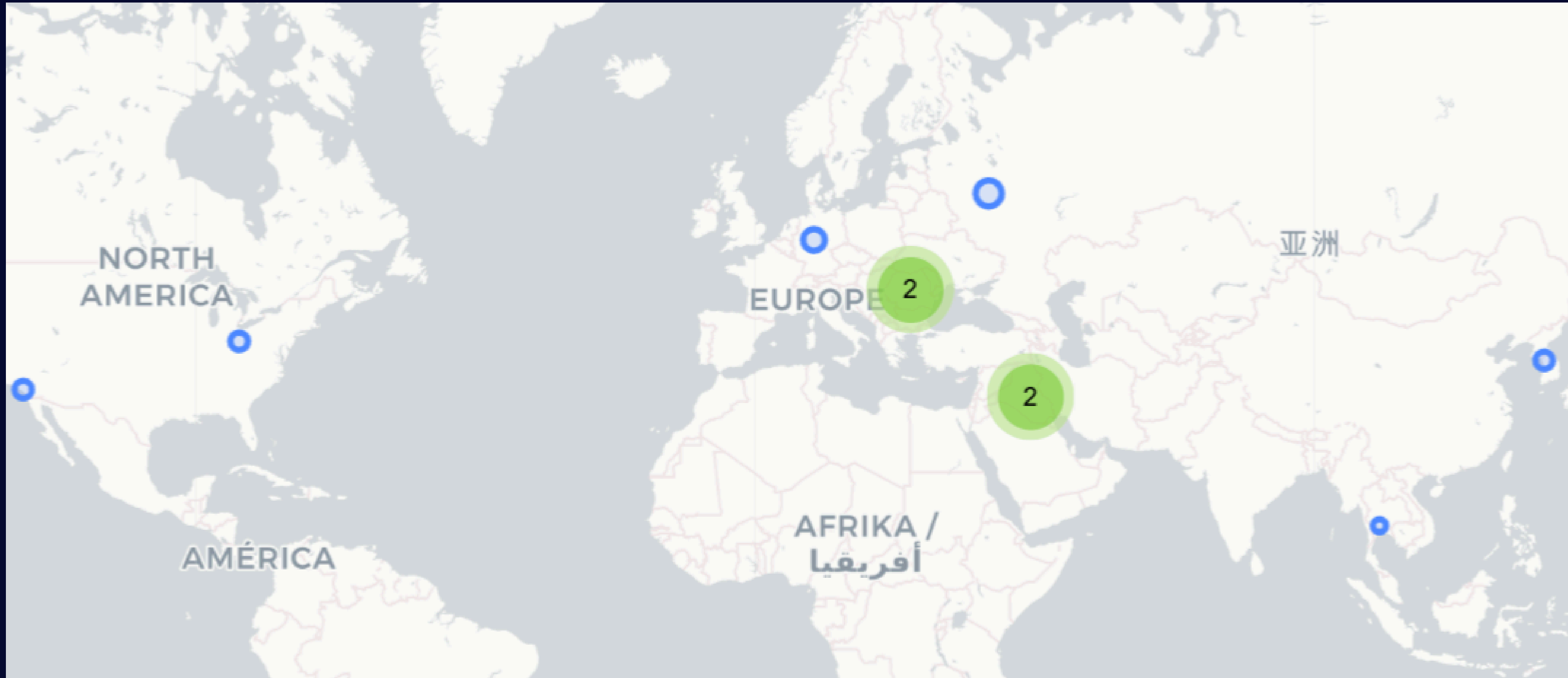
Azure VM, port 22 open, tcpdump
capture. Real traffic, not a sample.

```
azureuser@az900-vm1:~$ sudo pkill tcpdump
[sudo] password for azureuser:
azureuser@az900-vm1:~$ ls -lh /tmp/ssh_bruteforce.pcap
-rw-r--r-- 1 tcpdump tcpdump 968K Jun  1 18:19 /tmp/ssh_bruteforce.pcap
azureuser@az900-vm1:~$ scp azureuser@52.246.249.89:/tmp/ssh_bruteforce.pcap C:\Users\calip\Downloads\
```

SSH encrypts the password, not the handshake. libssh is an automated tool, not a person.

ip.src == 80.94.95.115 and ip.dst == 10.0.0.4 and tcp.dstport == 22 and frame contains "libssh"						
No.	Time	Source	Destination	Protocol	Length	Info
540	125.392714	80.94.95.115	10.0.0.4	SSHv2	89	Client: Protocol (SSH-2.0-libssh_0.10.5)
543	125.563928	80.94.95.115	10.0.0.4	SSHv2	1218	Client: Key Exchange Init

Pulled 9 unique IPs out of the capture and mapped them. Not one attacker, this is the constant background scanning every exposed server lives with.



Each one a separate machine. Connect,
fail, disconnect, repeat.

Wireshark · Endpoints · ssh_bruteforce.pcap

Endpoint Settings

- ☐ Name resolution
- ☐ Display raw data
- ☐ Hide aggregated
- ☒ Limit to display filter

Copy

Ethernet · 2	IPv4 · 10	IPv6	TCP · 10	UDP						
Address	Packets	Bytes	Total Packets	Percent Filtered	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes	Country	
10.0.0.4	178	37 kB	1,302	13.67%	82	17 kB	96	20 kB		
80.94.95.116	31	5 kB	31	100.00%	17	3 kB	14	3 kB		
80.94.95.115	28	5 kB	28	100.00%	14	2 kB	14	3 kB		
65.20.143.45	24	6 kB	24	100.00%	14	4 kB	10	3 kB		
65.20.217.64	22	6 kB	22	100.00%	13	3 kB	9	3 kB		
121.128.84.224	22	6 kB	22	100.00%	13	3 kB	9	3 kB		
183.89.250.247	22	6 kB	22	100.00%	13	3 kB	9	3 kB		
83.239.171.226	14	1 kB	14	100.00%	3	192 bytes	11	943 bytes		
88.214.25.124	8	558 bytes	8	100.00%	4	287 bytes	4	271 bytes		
3.22.234.7	7	521 bytes	7	100.00%	5	338 bytes	2	183 bytes		

The part that matters. Same IPs in the pcap AND my Sentinel logs. Two tools, one attack.

Home > Microsoft Sentinel

Microsoft Sentinel | Logs

Selected workspace: 'law-sentinel-lab-001'

New Query 1* ... +

Observability agent New Save ▾ Share ▾ ... | Queries hub

Run | ▾ Time range: Set in query Show: 1000 results KQL mode ▾

```
1 let AttackerIPs = dynamic([
2   "3.22.234.7",
3   "65.20.143.45",
4   "65.20.217.64",
5   "80.94.95.115",
6   "80.94.95.116",
7   "83.239.171.226",
8   "88.214.25.124",
9   "121.128.84.224",
10  "183.89.250.247"
11 ])
```

Results Chart | ☆ Add bookmark

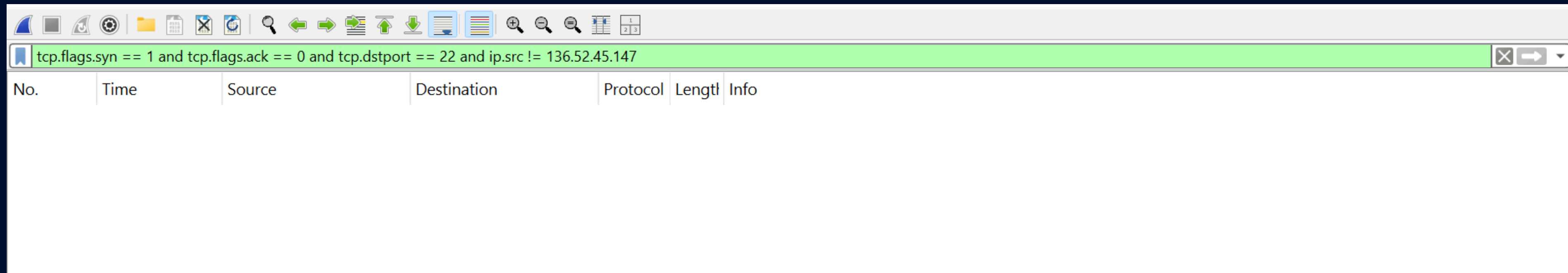
☐	TimeGenerated [UTC]	Computer	Facility	SeverityLevel	SyslogMessage
☐	> 6/1/2026, 1:42:19.888 PM	az900-vm1	auth	info	Connection closed by authenticating user sshd 80.94.95.116 port 3570..
☐	> 6/1/2026, 1:42:17.578 PM	az900-vm1	auth	info	Failed password for sshd from 80.94.95.116 port 35704 ssh2
☐	> 6/1/2026, 1:42:15.220 PM	az900-vm1	authpriv	notice	pam_unix(sshd:auth): authentication failure logname= uid=0 euid=0 tt..
☐	> 6/1/2026, 1:32:20.550 PM	az900-vm1	auth	info	Connection closed by invalid user operator 80.94.95.116 port 16770 [p..
☐	> 6/1/2026, 1:32:19.905 PM	az900-vm1	auth	info	Failed password for invalid user operator from 80.94.95.116 port 1677...
☐	> 6/1/2026, 1:32:17.978 PM	az900-vm1	authpriv	notice	pam_unix(sshd:auth): authentication failure logname= uid=0 euid=0 tt..

Columns

Every failed login leaves a mark in
the TCP flags. Connect, reset, gone.
Over and over.

tcp.port == 22 and ip.addr != 136.52.45.147						
No.	Time	Source	Destination	Protocol	Length	Info
111	177.839444	218.28.18.2	10.0.0.4	TCP	60	51157 → 22 [RST, ACK] Seq=1 Ack=1 Win=8192 Len=0
112	208.841629	218.28.18.2	10.0.0.4	TCP	60	51157 → 22 [RST, ACK] Seq=1 Ack=1 Win=8192 Len=0
113	240.835960	218.28.18.2	10.0.0.4	TCP	60	51157 → 22 [RST, ACK] Seq=1 Ack=1 Win=8192 Len=0

Locked the firewall to my IP, re-captured, filtered for outside attempts. Zero. The fix worked, and I proved it at the packet level.



The image shows a screenshot of the Wireshark network protocol analyzer. The top toolbar contains various icons for file operations, navigation, and analysis. Below the toolbar, a green filter bar contains the text: `tcp.flags.syn == 1 and tcp.flags.ack == 0 and tcp.dstport == 22 and ip.src != 136.52.45.147`. Below the filter bar is a table with the following columns: No., Time, Source, Destination, Protocol, Length, and Info. The table is currently empty, indicating that no packets matching the filter were captured.

No.	Time	Source	Destination	Protocol	Length	Info
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SSH source restricted from Any to just my IP.

Everything else dropped at the Azure edge before it

reaches the VM.

Network security group **az900-vm1-nsg** (attached to networkInterface: az900-vm1829)
Impacts 0 subnets, 1 network interfaces

Search rules

Source == all Destination == all Protocol == all Action == all Port == all

Prio... ↑	Name		Port	Protocol	Source	Destination	Action
Inbound port rules (5)							
300	SSH		22	TCP	136.52.45.147/32	Any	✓
310	Allow-8080-test		8080	TCP	Any	Any	✓
65000	AllowVnetInBound ⓘ		Any	Any	VirtualNetwork	VirtualNetwork	✓
65001	AllowAzureLoadBalancerInB... ⓘ		Any	Any	AzureLoadBalancer	Any	✓
65500	DenyAllInBound ⓘ		Any	Any	Any	Any	✗
> Outbound port rules (3)							

Caught it. Analyzed it. Matched it to my logs. Locked it down. Proved it.

That's MITRE T1110 — the industry code for brute force. The packet layer and the SIEM layer are both the job.

Next lab: Active Directory attack detection. The hands-on learning continues.